

1 Hypothesis

Will the addition of mechanics that allow the player to traverse the map in different ways allow the use of bigger maps or will it cause the player to be frustrated with the map despite the extra mechanics to traverse the map.

2 Player Actions

2.1 General Mechanics

The general actions the player can take are moving, sprinting, crouching toggling a player light and general interactions. The player cannot sprint indefinitely and can only sprint for 1 second where their speed is doubled for that 1 second. When the player moves they make a footstep sound every 5 seconds. When the player is sprinting they make footstep sounds every 0.5 seconds. When the player is crouching they do not make footstep sounds. The footstep sounds do nothing while in the first level of the game which is the narrator level, while the footstep sounds alert the enemy to the player's position in the procedurally generated level. This is a mechanic that changes based on the player's actions. Toggling the player light helps show hidden paths the player can take but it does take away the player's vision of what is around them the hidden paths are one of the Hypothesis mechanics. Having the light off also helps the player hide from the enemy, but will remove the warning that the player light gives when the enemy is nearby. The player can only interact with one interaction at a time and if the interaction needs the player to remember something or input something the interaction does not freeze the in game time to give the player a sense of urgency. One type of player interactions will be teleporters which are part of the Hypothesis mechanics.

2.2 General Challenges

Most of the challenges appear in the procedural generation level of the game, due to the narrative level being more of an exploration for the player where the player is led through the level. The main challenge in the procedural generation level is hiding and running away from the enemy. The enemy's speed is faster than the player's speed, but slower than the player's sprint speed. This brings another challenge where the player needs to manage their sprint in order to be able to escape from the enemy. The player will a

3 Goals

General Goal List:

- Connect all the prototypes.
- Add multiple ways to traverse the procedurally generated map.

4 General design outline

4.1 Story

The player will find themselves in a sort of laboratory where a disembodied voice will try to guide the player out of the laboratory. If the player follows the voice they will successfully escape the laboratory. If the player disobeys the voice they will be taken deeper into the laboratory where they will then have to escape from the laboratory without the help of the narrator.

4.2 Game Progression

The general game progression is that the player will start in the tutorial and then make their way from the first level to the second level. The first level will be handmade and will have a narrator. The second level will be a procedurally generated level.

4.3 Tutorial

The player will start in the tutorial where they will learn the basics of the game. The player will first learn the basic controls which are: Moving, sprinting, toggling the player light and general interactions the player can perform. The player will be introduced to the enemy in the tutorial and different ways to avoid them and how the enemy generally works.

4.4 First Level

The player will be put in a laboratory map where a disembodied voice called the narrator will then try to help the player escape from the laboratory. If the player follows the narrator then they will escape from the laboratory, if the player disobeys the narrator they will go deeper into the laboratory. The enemy will not appear in this level. This level will not be procedurally generated. This is to explore how the different mechanics shape the player experience in traversing the map.

4.5 Second Level

The player will be in a laboratory that is decayed. The player will face an enemy that is constantly trying to catch them. The player will also need to try and solve some sort of puzzle which could be finding a code or bringing an object from point A to point B. The decay is shown by giving the procedural generation a high level of random walk.

5 Genre

The genre of the game will be a stealth survival horror. A game that has these genres is Alien: Isolation

6 Possible Puzzles

These are the puzzles that can appear during the procedural generation of the level. Each puzzle will have success criteria which is determined by the puzzle.

6.1 Fetch quest

The player will need to find an item and bring it to the destination.
The generation will be a success if:

- The item can be accessed.
- The item can be picked up and put down.
- The destination can be accessed.
- The destination recognizes the item.
- The player completes the goal after bringing the item to the destination.

6.2 Password

The player will need to find a password that they will then use to open a lock.
The generation will be a success if:

- All parts of the password can be accessed
- All parts of the password can be read.
- The lock can be accessed.
- The lock successfully recognizes the entered password.
- The lock completes the goal after the password has been entered.

7 Prototype Improvements

This prototype will improve upon all previous prototypes, while connecting all the prototypes together.

7.1 Prototype 1 (Narrative Level) Improvements

The map of prototype 1 will be improved while also allowing the player to use more of the mechanics to explore the map. These improvements include making the map completely rigid compared to the current prototype 1 where half the map is rigid while the other half isn't. This will be done to more accurately communicate that the player is in a laboratory. This will include areas that can only be accessed when the player's light is turned off and areas that have a timer that the player needs to sprint towards to access. This will give the player more interaction in exploring the map.

7.2 Prototype 2 (Tutorial) Improvements

The map of prototype 2 will be improved to make the tutorial a more optional experience instead of a forced experience, this will be done by taking the linear map of the current tutorial and changing it to allow the player to learn any mechanic in any order. It will gradually show the player the mechanics without overwhelming them by showing them all the mechanics at once. A small tutorial will be shown to the player in the pause ui so the player can be reminded about the controls at any point during the game.

7.3 Prototype 3 (Enemy AI) Improvements

The only improvement being added to the Enemy AI is a static state machine state. This state will be used to allow the player to learn more about the enemy while in the tutorial. The state will cause the enemy to stand still while only listening to move commands from triggers, and the player's footsteps.

7.4 Prototype 4 (Procedural Generation) Improvements

The procedural generation algorithm will be improved by keeping track of all the rooms in the generation to allow adding of special rooms which will either be a room in which a puzzle needs to be solved or will have an interaction for the player. The algorithm will also allow for secret passages between rooms.

8 Communication Design

8.1 Mechanics

The sprint mechanic is communicated through zooming out the player's camera a bit to add the indication of speed, while also draining a bar on the left side of the screen to indicate that the player cannot sprint indefinitely. The crouch mechanic is communicated through zooming in the player's camera and by slowing the player's speed while they are crouching. The enemy warning is communicated by turning the player's light yellow when the enemy is nearby, while hidden paths that show when the light is turned off keep the blue colour

of the ground and are not affected by any light to emphasize the paths to the player. A prompt appears on the player's screen telling the player

8.2 Prototype 1 (Narrative level)

8.3 Prototype 2 (Tutorial)

Each mechanic is explained to the player through text. The player can then use the mechanics the way the text explains it to further learn the mechanics. Each mechanic has its own area that the player can accessed by stepping through teleporters to take the enemy to that mechanic's area.

8.4 Prototype 3 (Enemy AI)

The enemy has a red light around it to indicate that it is dangerous. Heavy footsteps also play whenever the enemy moves to a new destination. The sound does not play when the enemy moves, but only when the enemy pathfinding finds a new spot for the enemy to move to. When the enemy spots the player a red tint covers the screen along with the words "RUN" to indicate that the player is being chased and to add tension.

8.5 Prototype 4 (Procedural Generation)

To indicate that the enemy knows about the player's position a radar ping sound (reference sound) plays every few seconds when using the minimap. Each intractable is highlighted with a green light (change colours to be the same) this shows the player that they can interact with anything in a green light. A popup also appears prompting the user to press space which also helps the player know they can interact with specific game elements.

9 Process

9.1 Tutorial Design - Prototype 2

The design philosophy for the tutorial is that it needs to introduce the player to the general mechanics. This means it first needs to show the player how to move and then after confirming that the player knows how to move will then allow the player to select which mechanic they want to learn about. This allows the player freedom in the tutorial

9.2 Prototype 1

No timer areas were added to the level because the first level is made to feel like the player is calmly being guided through the level while the timer areas will just add unnecessary anxiety to the player.

9.3 Prototype 3

Fixed a bug in which caused the enemy to move as if idle when being sent to specific spots on the map. This bug sometimes stopped the enemy from moving towards the player when using the minimap causing the minimap to be less risky to use. The map made for prototype 3 was just to show off the enemy thus it has been removed making the final level progression of the game be: Prototype 2, Prototype 1, Prototype 4.

9.4 Prototype 4

Increased the player light size because the original light size made the player feel too cramped in the room causing the player to get lost in the map a bit too often. The original size was a 4 unit circle while the new size is a 7 unit circle. The detection for the enemy to see the player light is still the original size to allow the player some leeway in not getting spotted by the enemy but still being able to see what is happening. A minimap was added to the game to help to give the player a sense of direction, because players got confused with the maze-like map structure. The minimap will ping the players position every few seconds which will guide the enemy to the player. This is to make the minimap a risk for the player to use for too long while still helping the player navigate the level.

Text was added to the puzzle solution to help the player remember which parts of the solution the player has already found. This was done because the players would usually forget the solution parts they have found because the enemy chasing them causes the players to forget what they just found in order to try escape the enemy.

10 Hypothesis reflection

10.1 Mechanics as story

During the first level the player is guided through the laboratory. There are these black spots that can sometimes block the player's path where the player will then need to find a way around them. This is where the player will use some of the movement mechanics added. These include turning off the player light to find secret paths and going to teleporters to teleport around the obstacle. Here the movement mechanics must be used to progress instead of just to traverse the map or run away. These mechanics show the player that the map can be traversed in different ways, but the player needs to use these traversals to continue the story. These mechanics in this situation allows for a bigger map because of the narrator which is guiding the player making the player not feel frustrated as they traverse the map, because they are being led through the map and are not in any danger. This can feel boring to players due to the games minimal graphics, but if the graphics were improved the boring areas can be turned into areas where environmental storytelling keeps the players attention.

10.2 Mechanics in procedural generation

The main room connections for the procedural generation algorithm only allows for the rooms to connect in one direction causing rooms to typically only have one entrance and exit. This causes some of the adjacent rooms to not be connected to each other. The added movement mechanics of secret paths allow the player to go to these adjacent rooms. The secret paths shorten the time a player would take to get from one room to another while allowing the player to actively feel like they are creating these shortcuts. Due to the shortcuts being made from the player toggling their light it does allow for an escape from the enemy when the player turns off their light to hide from the enemy. The teleporters also teleport the player from one side of the map to the opposite side of the map. They help the player traverse the map but also help the player escape from the enemy. The only risk they have is that you don't know what is on the other side of the teleporter which can cause the player to instantly be caught by the enemy.

These mechanics in this situation only allow for a bigger map if they are placed in the procedural generation frequently but not too frequent, this is because of the enemy chasing the player. The mechanics would allow for the player to be able to escape from the chase but will still need the player to try to hide from the enemy instead of relying on the mechanics. This will keep the tense atmosphere of the enemy chasing the player while allowing for the player to escape if they are prepared for the enemy. In this situation the minimum graphics does not take away from the experience, because the player is already constantly busy with an objective and is constantly wary of the enemy. A graphics improvement will help with the horror themes of the game.

11 Reflections

11.1 Prototype 1

In the original prototype 1 the main focus was on the dialog system where the dialog system would try to keep the player invested in the game by telling a story that the player will be curious about. This prototype only allowed the player to move which was the only mechanic. Where the player moved was decided by if the player followed the narrator or not. The improved prototype 1 still keeps the dialog system that tells the player a story and guides the player through the level where the player can decide if they want to listen to the dialog system or not. Some improvements that were made upon the prototype are the inclusion of the mechanics in the game. These include toggling the player light to get paths certain areas or using the teleporter to progress, while also needing to pick up an item and move it to another area. These inclusion of mechanics help the player learn the mechanics and make the level more interesting instead of listening or not listening to the dialog system, because the player is actively moving through the level using other mechanics instead of just moving. The level was also changed from being rigid in areas you are supposed to be and

random in areas you are not supposed to be to rigid everywhere. Black blocks were added to indicate where you should not go. These map changes help the player understand more from where they should and shouldn't be according to the story while not confusing the player about the map design. (REDO)

11.2 Prototype 2

The original prototype 2 lead the player through the tutorial by introducing the mechanics one by one forcing the player to learn the first mechanic before being able to learn more mechanics. This was really restrictive, thus the newer prototype tutorial allows the player to learn any mechanic whenever they want after they learnt how to move around. This allows the player to quickly look at a mechanic that they want to know more about instead of having to go through the entire tutorial again. This improvement gives the player more freedom in what they want to learn about the game and when they want to learn about it.

11.3 Prototype 3

There was a bug with the enemy when the director or minimap tried to move the enemy closer or further away from the player, but the idle enemy state overwrote that action and moved the enemy around itself. This bug did sometimes make the enemy feel as if it was hunting the player when the player was hiding in a hiding spot. Causing the enemy to look like they are looking around the hiding spot for the player. When the enemy did this it was inconsistent, but it needed to be changed because of the addition of the minimap mechanic. The enemy did not constantly move towards the player when the minimap was active, because of this bug the minimap did not have the intended risk of use where the player finds out more about their surrounding but causes the enemy to go to their current position. Thus the bug had to be removed. The bug could be turned into a feature, but only when the enemy has lost sight of the player or when searching around a hiding spot where the player may be. This feature will improve the enemy and make it feel like its hunting the player.

11.4 Prototype 4

The original prototype 4 was only the procedural generation with an enemy and a fetch quest puzzle. The improved prototype added more mechanics to the procedural generation allowing the player to find hidden paths to move around and allowing the player to easily get to the other side of the map. The improved prototype is more interactive with the player where the player needs to actively search for the puzzle parts and actively activate the secret paths. These interaction improved the players enjoyment of the game. A minimap was added to help the player navigate, but this minimap took away from the game by giving the player a sense of direction. This takes away from the horror elements of the game because the player feels a sense of dread because they are lost, which the minimap diminishes. The current procedural generation

generates the map in a maze-like way which may be too confusing for players, thus the minimap helps the players traverse the map. Both improvements want to be kept thus a solution was found. The solution is that if the player uses the minimap the enemy will be alerted to their position and will constantly go to the position of the player. This solution makes the minimap a limited resource that the player can use at a risk. This leads to the horror elements of the game staying by letting the player feel like they are in danger because of the use of the minimap, but it also helps the player traverse the maze-like map reducing the player's frustration.

12 references (delete subsections later)

12.1 Tutorials

<https://medium.com/@doandaniel/gamedev-protips-a-few-helpful-tips-on-crafting-a-better-tutorial-ffcc636039d3> <https://www.gamedeveloper.com/business/gamedev-thoughts-how-to-design-a-good-tutorial-for-your-indie-game>

12.2 Minimap

<https://pixabay.com/sound-effects/film-special-effects-radar-ping-306440/>

12.3 Genre